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Operations Research

Module: IN0024 **Date:** Friday 7th August, 2026
Examiner: Prof. Dr. Martin Bichler **Exam:** Endterm

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Σ (Endterm)	Grade (Endterm)	Grade intervall	Grade bonus	Final grade
38.0	4.7	[35.0; 41.0)	37.0	4.7

Notes:

- Please make sure that the total amount of credits stated above is correct.
- Note that a potential bonus is only considered if the final exam (without bonus) is passed.
- Solely the second correction (green color) is decisive.

Corrections:

The table below lists all corrections (image recognition and complaints during review) that are already considered in the calculation of your grade. If a problem or subproblem is listed multiple times, the correction with the highest number (column "Correction") takes precedence.

Problem	Correction	credits	Annotations
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School of Computation, Information and Technology
Technical University of Munich

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Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

Operations Research

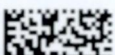
Exam: IN0024 / Endterm
Examiner: Prof. Dr. Martin Bichler

Date: Friday 7th August, 2026
Time: 17:00 – 19:00

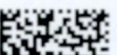
Working instructions

- This exam consists of **20 pages** with a total of **7 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 121 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one **non-programmable pocket calculator**
 - one **analog dictionary** English ↔ native language
- Subproblems marked by * can be solved without results of previous subproblems.
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from _____ to _____ / Early submission at _____



☐ Exam empty



Problem 1 Multiple Choice Questions (16 credits)

Mark all correct answers (there can be more than one):

Scoring: Each question is worth up to 2 points. You receive 2 points if you select all correct options and no incorrect ones. Selecting one incorrect option or not selecting a correct option reduces the score to 1 point. Any greater deviation results in 0 points.

Mark correct answers with a cross



To undo a cross, completely fill out the answer option



To re-mark an option, use a human-readable marking



0.0 a) Which of the following statements are correct?

- ☒ Consider $\min\{c^T x : Ax \leq b, x \in \mathbb{Z}^n\}$. If A is totally unimodular, then the optimal solution can be found in polynomial time.
- ☐ Every linear program can be solved in polynomial time.
- ☐ The branch-and-bound algorithm is guaranteed to find an optimal solution to an integer program in polynomial time.
- ☐ The simplex algorithm always finds an optimal solution to a linear program in polynomial time.

0.0 b) Which of the following statements are correct?

- ☐ If both the primal and dual problems are feasible, then they both have optimal solutions.
- ☒ If the primal problem is unbounded, then the dual problem is infeasible.
- ☐ If the dual problem is unbounded, then the primal problem is infeasible.
- ☐ If the feasible region is unbounded, then any linear program defined on it has an unbounded objective value.

2.0 c) Consider a maximization integer linear program solved using the Branch-and-Bound algorithm. Which of the following statements are correct?

- ☐ Branch-and-Bound always explores every feasible integer solution before proving optimality.
- ☒ If the LP relaxation at a node has an integer-optimal solution, then that solution is optimal for the corresponding subproblem.
- ☒ A node can be pruned if the value of its LP relaxation is worse than the value of the current incumbent solution.
- ☒ For a maximization problem, the objective value of the LP relaxation provides an upper bound on the objective value achievable by any integer-feasible solution in that node.

0.0 d) Consider a pure integer linear program whose LP relaxation has been solved optimally. The optimal LP solution is fractional. Which of the following statements are correct?

- ☒ If a Gomory cut is added to the LP relaxation, the optimal objective value of the LP relaxation can only improve (increase) for a maximization problem.
- ☒ A Gomory cut can be derived from a row of the optimal simplex tableau whose basic variable has a fractional value.
- ☐ A Gomory cut removes the current fractional LP-optimal solution but does not remove any feasible integer solution.
- ☐ A Gomory cut can be generated from the objective row of the simplex tableau.



0.0 e) Which statements about total unimodularity are correct?

- ☐ Every square matrix with determinant 1 is totally unimodular.
- ☒ A matrix containing an entry 2 cannot be totally unimodular.
- ☐ In a totally unimodular matrix, every square submatrix has determinant in $\{-1, 0, 1\}$.
- ☒ If A is totally unimodular and b is integral, then every feasible point of $\{x : Ax \leq b\}$ is integral.

1.0 f) A 1.5-approximation algorithm for a metric TSP instance returns a tour of cost 30. Let OPT denote the cost of an optimal tour. Which conclusions are guaranteed?

- ☐ $OPT = 20$.
- ☒ $OPT \geq 20$.
- ☐ The returned tour must be optimal.
- ☐ $OPT \leq 30$.

0.0 g) Which of the following statements are correct?

- ☐ If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex, then every stationary point of f is a local maximum.
- ☐ If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and differentiable, then every local minimizer of f is also a global minimizer.
- ☒ If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is strictly convex, then it always has a minimizer.
- ☒ If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice continuously differentiable and $\nabla^2 f(x) \succeq 0$ for all x , then f is convex.

0.0 h) Which of the following statements are correct?

- ☐ A KKT point of a non-convex optimization problem is always a global optimum.
- ☒ For a convex minimization problem satisfying Slater's condition, the KKT conditions are necessary for optimality.
- ☒ Slater's condition requires that all feasible points satisfy all inequality constraints strictly.
- ☐ For a convex minimization problem, any feasible point satisfying the KKT conditions is a global minimizer.

Additional space for notes:



Problem 2 Glass Production (21 credits)

A manufacturer develops specialty glasses. The glass is produced by mixing different oxides. The composition influences the physical properties of the glass as well as its production cost. Let M denote the set of available raw materials

$$M = \{\text{SiO}_2, \text{Li}_2\text{O}, \text{Na}_2\text{O}, \text{K}_2\text{O}, \text{MgO}, \text{CaO}, \text{SrO}, \text{BaO}, \text{B}_2\text{O}_3, \text{Al}_2\text{O}_3\}.$$

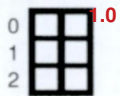
For every material $i \in M$, $x_i \in [0, 1]$ denotes its mass fraction in the final glass. Furthermore, $y_i \in \{0, 1\}$ indicates whether material i is used. The unit cost of material i is denoted by c_i . The customer specifies acceptable ranges for several material properties:

$$P = \{\text{conductivity, thermal expansion, refractive index, density, opacity}\}$$

For every property $k \in P$, a linear model predicts the value of property k given the glass composition:

$$P_k(x) = \sum_{i \in M} a_{ki} x_i.$$

Formulate the optimization problem mathematically using only linear constraints.



a) The objective is to minimize production cost.

$$\text{Min } \sum_{i \in M} x_i \cdot y_i \cdot c_i \quad x_i, y_i \text{ not linear}$$



b) The total composition must sum to 100%.

$$\sum_{i \in M} x_i \cdot y_i = 1$$



c) Silicon dioxide (SiO_2) forms the glass network. For mechanical stability, at least 60% of the glass must consist of SiO_2 .

$$x_s \cdot y_s \geq 0.6 \quad \text{with } s \text{ correspond to } \text{SiO}_2$$



d) Every selected material must contribute at least 2%.

$$x_i \geq 0.02 \cdot y_i \quad \forall i \in M$$



e) The produced glass needs to satisfy the property specifications defined by lower bounds L_k and upper bounds U_k for all properties.

$$\forall k: P_k(x) \geq L_k \quad \forall k: P_k(x) \leq U_k$$



f) Due to an existing patent, lithium oxide (Li_2O) and potassium oxide (K_2O) may not be used simultaneously.

$$y_l + y_k \leq 1 \quad \text{with } l \text{ correspond to } \text{Li}_2\text{O} \\ k \text{ correspond to } \text{K}_2\text{O}$$



- 2.0 g) Barium oxide (BaO) increases the refractive index of the glass but reduces its chemical durability. Therefore, barium oxide needs to be combined with aluminum oxide (Al_2O_3), to stabilize the glass network. Consequently, whenever BaO is used, Al_2O_3 must also be present.



$$y_a \geq y_b, \quad \text{with } a \text{ correspond to } \text{Al}_2\text{O}_3 \\ b \text{ correspond to } \text{BaO}$$

- 2.0 h) At most six different raw materials may be used.



$$\sum_{i \in M} y_i \leq 6$$

- 0.0 i) Given an integer variable $z \in \{0, 1\}$ and a variable $x \in [0, 1]$ model the product zx linearly using a new variable w .



$$\text{Let } w = xz. \quad w \leq x \\ w \leq z. \quad w \geq 0.$$

- 1.0 j) *Boron anomaly:*



The coefficient of thermal expansion is increasing for small concentrations of boron oxide (B_2O_3), but changes significantly once the concentration exceeds a threshold τ . Consequently, the thermal expansion can no longer be described by a single linear function.

Incorporate this effect into the optimization model.

Hint: introduce a binary variable $z \in \{0, 1\}$ that captures whether the concentration of B_2O_3 is above, or below the threshold τ and enforce this behavior suitably using linear constraints.

Use the z to define the new thermal expansion model $P_{\text{exp}}(x, z)$.

You can use the previous subproblem without proof to eliminate products of z and x_i .

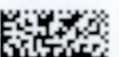
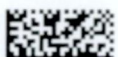
Let $z \in \{0, 1\}$, denotes whether the concentration of B_2O_3 is above or below τ , with 1 means above.

x_b denote mass fraction of B_2O_3

$$\begin{cases} z \geq 0 \\ z \leq (x_b - \tau)M \end{cases}$$

$$P_{\text{exp}}(x, z) = (1-z) \sum_{i \in M} a_{ki} x_i + z \cdot \left(\sum_{i \in M} a_{ki} x_i + M \right)$$

LP minimize



Problem 3 Duality (15 credits)

Consider the following primal linear program (P):

$$\begin{aligned}
 \max \quad & 4x_1 + 2x_2 + 3x_3 \\
 \text{s.t.} \quad & x_1 + x_2 + 2x_3 \leq 8 \quad S \\
 & 2x_1 - x_2 + x_3 \geq 2 \quad B \\
 & x_1 + x_2 = 4 \quad O \\
 & x_1, x_2, x_3 \geq 0. \quad \text{SSS}
 \end{aligned}$$

7.0 a) Derive the dual linear program (D) corresponding to (P).

$$\min \quad 8y_1 + 2y_2 + 4y_3$$

$$\text{s.t.} \quad y_1 + 2y_2 + y_3 \geq 4$$

$$y_1 - y_2 + y_3 \geq 2$$

$$2y_1 + y_2 \geq 3$$

$$\cancel{y_1 \geq 0} \quad y_1 \geq 0 \quad y_2 \leq 0 \quad y_3 \in \mathbb{R} \quad \checkmark$$



4.0 b) State the primal and the dual complementary slackness conditions explicitly.

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	4

primal CS conditions:

$$(x_1 + x_2 + 2x_3 - 8)y_1 = 0$$

$$(2x_1 - x_2 + x_3 - 2)y_2 = 0$$

$$(x_1 + x_2 - 4)y_3 = 0$$

dual CS conditions:

$$(y_1 + 2y_2 + y_3 - 4)x_1 = 0$$

$$(y_1 - y_2 + y_3 - 2)x_2 = 0$$

$$(2y_1 + y_2 - 3)x_3 = 0$$



4.0 c) Consider the feasible primal solution

$$x^* = (2, 2, 0)^T$$

and the feasible dual solution

$$y^* = (2, 0, 2)^T.$$

Can these two solutions both be optimal? Justify your answer.

	0
	1
	2
	3
	4

Let's say x^* is optimal. Replace in primal CS conditions:

$$(2 + 2 + 2 \cdot 0 - 8)y_1 = 0$$

$$(2 \cdot 2 - 2 + 0 - 2)y_2 = 0 \Leftrightarrow$$

$$(2 + 2 - 4)y_3 = 0$$

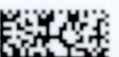
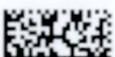
$$-4y_1 = 0 \checkmark$$

$$0 \cdot y_2 = 0$$

$$0 \cdot y_3 = 0$$

$\Rightarrow y_1$ must be 0 $\Rightarrow$ ~~contradict~~ contradict with y^*

$\Rightarrow$ the two solution can not both be optimal

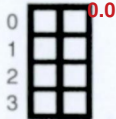


Problem 4 Branch and Bound (15 credits)

We consider the following knapsack problem with free variables:

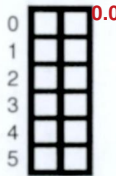
$$\begin{aligned} \max_x \quad & \sum_i v_i x_i = 20x_1 + 12x_2 + 2x_3 + 9x_4 \\ \text{s.t.} \quad & \sum_i w_i x_i = 2x_1 + 2x_2 + x_3 + 2x_4 \leq c = 5. \end{aligned} \quad (\mathcal{P}_{\text{free}})$$

The above model is denoted by $\mathcal{P}_{\text{free}}$, in which the variables can take any negative or positive real value. We also consider the same problem when the variables are restricted to be non-negative reals, non-negative integers, or binary values, and denote them by $\mathcal{P}_{\geq 0}$, $\mathcal{P}_{\mathbb{Z}_+}$, and $\mathcal{P}_{\{0,1\}}$, respectively. We want to solve these problems using the Branch and Bound algorithm.

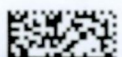
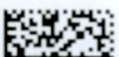


a) Are the problems $\mathcal{P}_{\text{free}}$ and $\mathcal{P}_{\geq 0}$ useful in solving the integer valued programs? Give your reasoning.

No, because the values are reals, not integer, so we can not bound the values range.



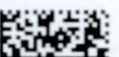
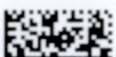
b) Apply the Branch and Bound algorithm for solving $\mathcal{P}_{\mathbb{Z}_+}$. Only iterate the first branch and solve the resulting subproblems. Also explain how the algorithm would further branch out without solving the resulting subproblems. You do not need to find the optimal solution. Hint: The LP relaxation of $\mathcal{P}_{\mathbb{Z}_+}$ has the following optimal solution, which you may use without justification: Determine the value-weight ratios $q_i = v_i / w_i$ and index the items such that $q_i \geq q_{i+1}$ for all $j = 1, \dots, n-1$. Set $x_k = c/w_k$ for $k = 1$, and $x_j = 0$ for $j > 1$. When further constraints are imposed, the optimal solution greedily takes items in decreasing order of their value-to-weight ratio until the capacity is full.





- 0.0 c) Solve $\mathcal{P}_{\{0,1\}}$ using the Branch and Bound algorithm. *Hint: The LP relaxation of $\mathcal{P}_{\{0,1\}}$ has the following optimal solution, which you may use without justification: Determine the value-weight ratios $q_i = v_i/w_i$ and index the items such that $q_j \geq q_{j+1}$ for all $j = 1, \dots, n-1$. Choose k such that $\sum_{j=1}^k w_j \leq c$, but $\sum_{j=1}^{k+1} w_j > c$. Set $x_j = 1$ for $j = 1, \dots, k$, $x_{k+1} = (c - \sum_{j=1}^k w_j)/w_{k+1}$, and $x_j = 0$ for $j > k+1$. When further constraints are imposed, the optimal solution greedily takes items in decreasing order of their value-to-weight ratio until the capacity is full.*

	0
	1
	2
	3
	4
	5
	6
	7



Problem 5 Gomory Cut (15 credits)

Consider the following integer programme:

$$\max z = 5x_1 + 8x_2 \quad (5.1)$$

$$\text{s.t. } x_1 + x_2 + s_1 = 6 \quad (5.2)$$

$$5x_1 + 9x_2 + s_2 = 45 \quad (5.3)$$

$$s_1, s_2 \geq 0 \quad (5.4)$$

$$x_1, x_2 \in \mathbb{N}_0 \quad (5.5)$$

After solving the LP relaxation with the simplex method, the following optimal tableau is obtained:

	x_1	x_2	s_1	s_2	RHS
z	0	0	$5/4$	$3/4$	$165/4$
x_1	1	0	$9/4$	$-1/4$	$9/4$
x_2	0	1	$-5/4$	$1/4$	$15/4$

- 2.0 a) Determine the LP-optimal solution and explain why it is not feasible for the integer program.

s_1, s_2 are slack vars $\Rightarrow$ they have value of zero
 $x_1 = 9/4$ optimal
 $x_2 = 15/4$ both values are non integer
 $\Rightarrow$ the optimal solution is not feasible for the IP

- 0.0 b) Derive a Gomory fractional cut from the row corresponding to the basic variable x_2 . Show all intermediate calculations.

$$x_2 \mid 0 \quad 1 \quad -5/4 \quad 1/4 \mid 15/4$$

~~$$s_2 \mid 0 \quad 0 \quad 1 \quad 0 \mid 0$$~~

~~$$x_2 \mid 0 \quad 1 \quad -5/4 \quad 1/4 \mid 15/4$$~~

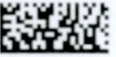
$$x_2 \mid 0 \quad 1 \quad -5/4 \quad 1/4 \mid 15/4$$

Cut $3/4 \mid 15/4 = 1/5 = 20\%$ to bring s_2 down to integer nearest

$$\rightarrow x_2 \mid 0 \quad 1 \quad -5/4 \quad 1/4 \mid 15/4$$

Multiply row by ~~5/4~~ $9/5$

$$x_2 \mid 0 \quad 1 \quad -1 \quad 1/5 \mid 3$$



0.0 c) Verify that the current LP-optimal solution violates the Gomory cut obtained in the previous part.

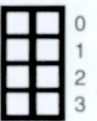


Handwritten solution for part c):

$$\begin{aligned} \max \quad & z = 5x_1 + 8x_2 \\ \text{s.t.} \quad & x_1 + 1.2x_2 + s_1 = 6 \\ & 5x_1 + 10.8x_2 + s_2 = 45 \end{aligned}$$

The optimal solution is $(5.4, 5.5)$.

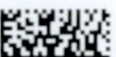
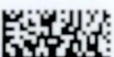
0.0 d) Formulate the LP relaxation that would be solved after adding the cut obtained in Part (2). You may reference original constraints (1)–(5) instead of rewriting them completely.



Handwritten solution for part d):

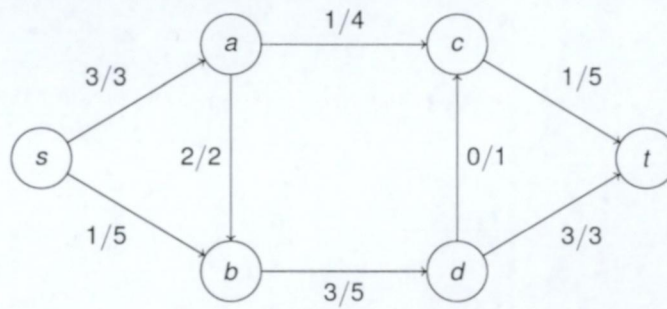
$$\begin{aligned} \max \quad & z = 5x_1 + 8x_2 \\ \text{s.t.} \quad & x_1 + 1.2x_2 + s_1 = 6 \\ & 5x_1 + 10.8x_2 + s_2 = 45 \end{aligned}$$

The optimal solution is $(5.4, 5.5)$.



Problem 6 Maximum Flow (17 credits)

Consider the following flow network. The label f/u on an arc denotes its current flow f and capacity u .



- 0 1.0 a) State the value of the current flow.

4 (3+1 out of s and 1+3 into t) ✓

- 0 1.0 b) Determine all arcs with positive capacity in the residual network and state their capacities. You may present your answer as a drawing or a list.

$s \rightarrow b : 4$

$a \rightarrow c : 3$

$b \rightarrow d : 2$

$d \rightarrow c : 1$

$c \rightarrow t : 4$

backward edges
missing



1.0 c) Starting from the given flow, run the Ford-Fulkerson algorithm until no augmenting s - t path remains. For every augmentation, state:

- the chosen s - t path in the current residual network;
- the amount by which the flow is augmented; and
- the resulting flow on every original arc whose flow changes.

	0
	1
	2
	3
	4
	5
	6
	7

Finally, state the value of the resulting flow.

① $s \xrightarrow{1/5} b \xrightarrow{3/5} d \xrightarrow{3/3} t$ f.
augment 3 from $s \rightarrow b$

$s \xrightarrow{1/1} b$

② $s \xrightarrow{1/1} b \xrightarrow{3/5} d \xrightarrow{3/3} t$ f.
augment 2 from $b \rightarrow d$

$b \xrightarrow{3/3} d$

③ $s \xrightarrow{3/5} a \xrightarrow{1/1} c \xrightarrow{1/5} t$ f.

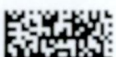
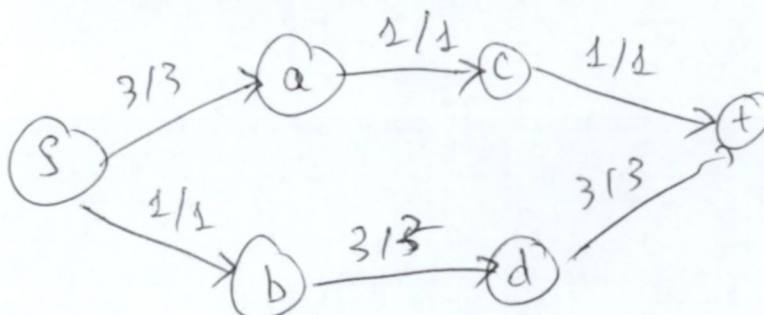
augment 3 from $a \rightarrow c$

$a \xrightarrow{1/1} c$

④ $s \xrightarrow{3/3} a \xrightarrow{1/1} c \xrightarrow{1/5} t$ f.

augment 4 from $c \rightarrow t$

$c \xrightarrow{1/1} t$



0
1
2
3

1.0 d) Find an s - t cut whose capacity equals the value of the resulting flow. State the two node sets defining the cut and compute its capacity. Explain why this proves that the resulting flow is maximum.

~~$\{s, a\}, \{b, c, d, t\}$~~

~~$s \rightarrow a$: 3 capacity~~

$\{s\}, \{a, b, c, d, t\}$

↳ capacity: $s \rightarrow a : 0$ $s \rightarrow b : 4$ $\Sigma = 4$ f.

this is the minimum cut

minimum cut $\nrightarrow$ maximum flow ✓

$\Rightarrow$ the resulting ~~max~~ flow is maximum ✓

0
1
2

0.0 e) Briefly justify why a maximum flow can be chosen integral when all capacities are integral.

✓



Problem 7 Budget Allocation (22 credits)

A company allocates one unit of budget between two advertising channels. Let $x_1 \in [0, 1]$ denote the fraction of the budget assigned to channel 1 and $x_2 \in [0, 1]$ the fraction assigned to channel 2. Due to diminishing returns, the resulting utility is modeled as

$$f_\alpha(x) = \log(1 + x_1) + \alpha \log(1 + x_2),$$

where $\alpha > 0$ describes the relative importance of channel 2 and $\log$ denotes the natural logarithm. The company wants to solve

$$(P_\alpha) \quad \max_{x \in \mathbb{R}^2} \log(1 + x_1) + \alpha \log(1 + x_2)$$

subject to

$$x_1 + x_2 \leq 1, \quad x_1 \geq 0, \quad x_2 \geq 0.$$

Equivalently, the inequality constraints can be written as

$$g_1(x) = x_1 + x_2 - 1 \leq 0, \quad g_2(x) = -x_1 \leq 0, \quad g_3(x) = -x_2 \leq 0.$$

- 0.0 a) State Slater's condition for a convex optimization problem with inequality constraints.

All feasible points satisfy all inequalities strictly.

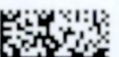
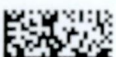
↯



- 0.0 b) Show that f_α is concave on the feasible set.

$$H_{f_\alpha} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \not\geq 0 \quad \times$$

$\Rightarrow f_\alpha$ is concave.



0.0 c) Show that the feasible set of (P_α) is convex and that Slater's condition holds.

$x_1 + x_2 \leq 1 \quad x_1 \geq 0 \quad x_2 \geq 0.$

$\forall x_1, x_2$ that satisfy the inequality:
 $x_1, x_2 \in [0, 1].$

Given $\lambda \in [0, 1], x, y$
 ~~$\lambda x_1 + (1-\lambda)x_2$~~
 ~~$\lambda x + (1-\lambda)y = \lambda x_1 + (1-\lambda)x_2$~~

$\lambda x_1 + (1-\lambda)x_2 = z$

~~feasible set of (P_α) is convex.~~

z also belongs to feasible set of (P_α)
 $\Rightarrow$ the set is convex.

z reaches min when $x_1, x_2 = 0$ since $\lambda, 1-\lambda \geq 0$
 min $z = 0$

z reaches max when $x_1, x_2 = 1$ max $z = \lambda + 1 - \lambda = 1$

1.0 d) In addition to the utility, the company accounts for an administrative overhead proportional to the larger of the two allocations,

$$c(x) = \max\{x_1, x_2\}.$$

State the definition of a convex function and show *directly from the definition* that c is convex on $\mathbb{R}^2$.

Definition:

A function f is convex when:
 given $\lambda \in [0, 1]$

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y). \quad \checkmark$$

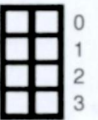
~~$f(\lambda(x_1+y_1) + (1-\lambda)(y_1+y_2))$~~

$\neq$ Proof in additional space!



In the following subproblems, set up the KKT conditions of (P_α) and determine the optimal solution x^* for arbitrary $\alpha > 0$, distinguishing cases for α . Work through the following steps.

- 0.0 e) Write (P_α) as a minimization problem, state the Lagrangian, and write down all KKT conditions.

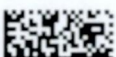
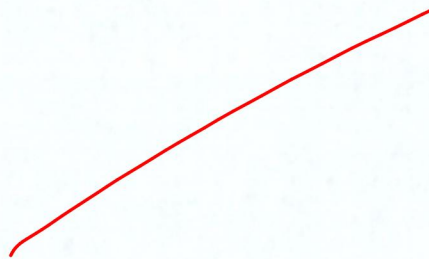


P_α as minimization problem:

$$\begin{array}{ll} \text{Min} & -\log(x_1+1) - \alpha \log(1+x_2) \quad (\checkmark) \\ \text{s.t} & x_1+x_2 \geq -1; \quad x_1 \leq 0 \quad x_2 \leq 0 \end{array}$$

Lagrangian, KKT-conditions?

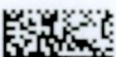
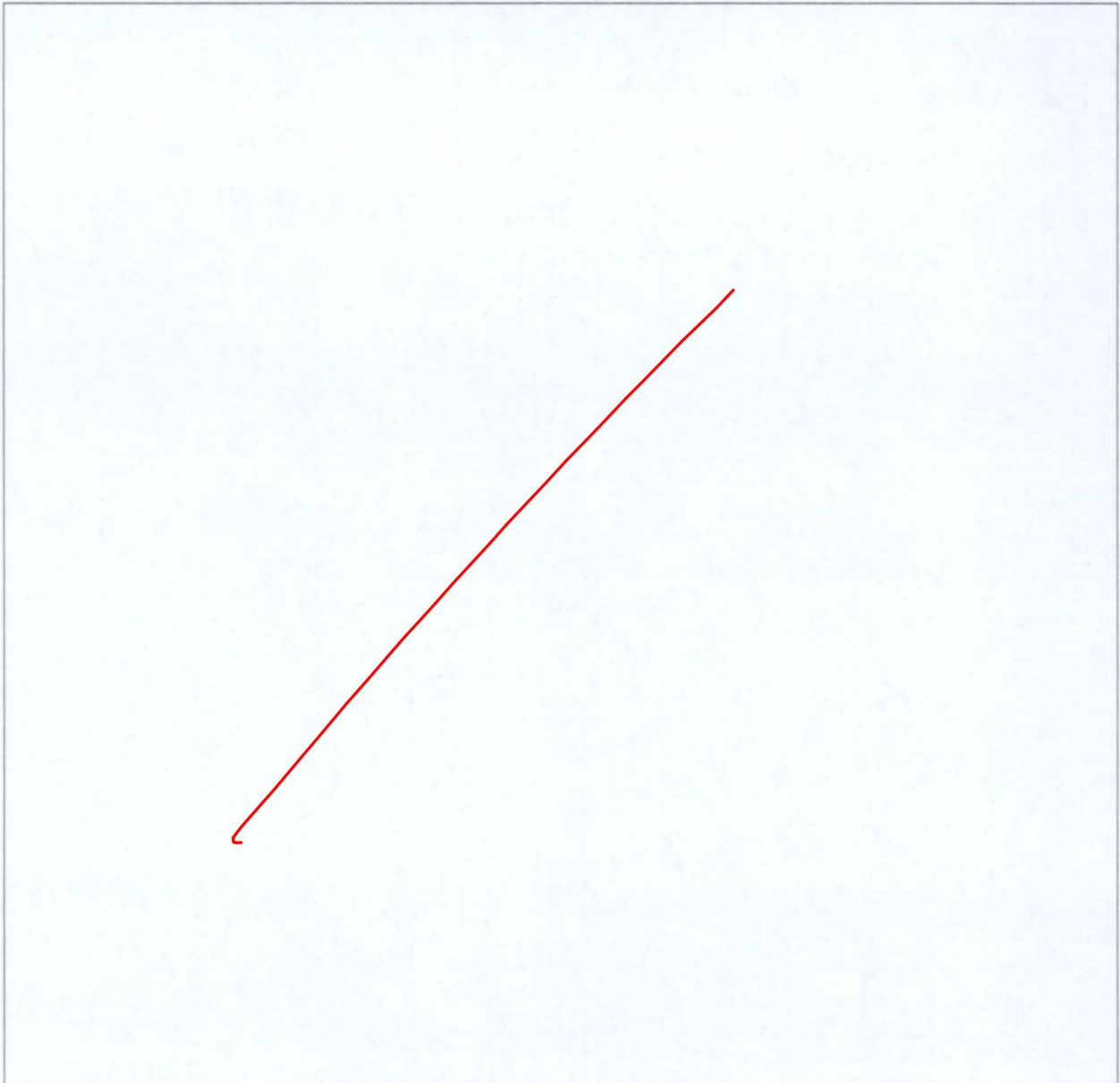
- 0.0 f) Show that the budget constraint is active at every KKT point, i.e. $x_1 + x_2 = 1$.





0	
1	
2	
3	
4	
5	
6	

g) Using $x_2 = 1 - x_1$ from (f), analyse the three cases $x_1, x_2 > 0$, $x^* = (1, 0)^T$, and $x^* = (0, 1)^T$. For each case determine the multipliers and the range of α for which the case is consistent with the KKT conditions. Summarize the result and justify why the KKT points are global maxima.





Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

~~7. b:~~

7. d

Let $x = (x_1, x_2)$ with $x_1 \geq x_2 \Rightarrow f(x) = x_1$

$y = (y_1, y_2)$ with $y_1 \geq y_2 \Rightarrow f(y) = y_1$

(1)

$$\cancel{f(\lambda x + (1-\lambda)y)}$$

$$= \cancel{f(\lambda x_1 + \lambda x_2 + (1-\lambda)y_1 + (1-\lambda)y_2)}$$

$$= \cancel{f(\lambda x_1 + (1-\lambda)y_1) + (\lambda x_2 + (1-\lambda)y_2)}$$

$$\text{Let } z = \max^2(\lambda x_1 + (1-\lambda)y_1, \lambda x_2 + (1-\lambda)y_2)$$

$$f(\lambda x + (1-\lambda)y)$$

$$= f(z)$$

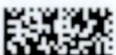
$$\text{Since } x_1 \geq x_2, y_1 \geq y_2 \Rightarrow \lambda x_1 + (1-\lambda)y_1 \geq \lambda x_2 + (1-\lambda)y_2$$

X

$$f(z) \leq \lambda x_1 + (1-\lambda)y_1 = \lambda f(x) + (1-\lambda)f(y) \quad (\text{from (1)})$$

$\Rightarrow f$ is convex.

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$$\log(1+x_1) + 2\log(1+x_2)$$

$$\max \quad 5x_1 + 3x_2$$

$$\text{s.t.} \quad x_1 \leq 3$$

$$x_2 \leq 4$$

$$x_1 + x_2 \leq 5$$

$$\min \quad -(5x_1 + 3x_2)$$

$$x_1 \geq -3$$

9/4

$$2 + \frac{1}{4} + \frac{15}{4} = 6$$

$$5 \quad 3 \quad 5.25$$